Participant Handbook

Foundations of Clinical Research · Level 1: From Clinical Question to Publication

Provider: Neudata · **#ClearDataClearImpact**

Welcome

Every day you make clinical decisions that research could improve. Over eight weeks, with one online session every Saturday evening (19:30 to 22:30, Vietnam time), this course takes you through the whole research process: from a question at the bedside, to a sound study design, to analysing data, to writing and publishing the paper.

You do not need any research or statistics background. We go slowly, with pictures, one running example and a lot of short polls and questions in the chat. The course is fully online and you join from home. There are no groups or breakout rooms: the whole class works through every exercise together, with the instructor. You will never be asked to calculate anything beyond a simple 2×2 table, and you do not need any software: the trainers run the analyses, and you learn to read and write them up.

The course at a glance

Week	Session (Saturdays, 19:30 to 22:30, online)	Theme	You build (homework)
1	1	**ASK:** from clinical problem to research question	Proposal: topic, PICO, question, aims
2	2	**FIND & READ:** searching, managing references, reading a paper in 15 minutes	Proposal: search, 5 references, problem statement, working title
3	3	**DESIGN:** choosing the right study	Proposal: design, setting, population
4	4	**BIAS & EFFECT:** bias, confounding, RR/OR/NNT, diagnostic tests, forest plots	Proposal: exposure, outcomes, bias plan
5	5	**DATA:** CRFs, data management, describing data, *meeting the course dataset*	Proposal: variables, CRF, data plan · Manuscript: Table 1 understood
6	6	**ANALYSE:** CIs, p-values, tests, regression, survival, missing data, *analysing the course dataset*	Proposal: sample size, analysis plan · Manuscript: results interpreted, question chosen

Week	Session (Saturdays, 19:30 to 22:30, online)	Theme	You build (homework)
7	7	**PROTECT & WRITE:** ethics, integrity, Methods and Results	Proposal: ethics, feasibility, publication plan · Manuscript: Results section
8	8	**PUBLISH:** Introduction, Discussion, abstract and title, improving the Results, journals, submission, peer review and presentations	Manuscript: Results section submitted · Proposal submitted (about 5 volunteers present)

Each session lasts 3 hours with a 15-minute break. It has a short recap, two teaching blocks with frequent "check your understanding" moments (a quick poll or an answer in the chat), and a proposal-builder segment in which the instructor shows the next part of the proposal on the running example. You then write the same part for your own question at home. Each session also has extra slides at the end for anyone who wants to read further at home.

The running example

One clinical topic travels with us: **fluids in septic shock**. In Weeks 1 to 4 we ask, search and design the question, and look at two real papers (a randomised trial and a cohort study). In Weeks 5 to 8 we read the results of FLUID-ICU, a realistic *simulated* dataset of 620 ICU patients, and you report them as a Results section.

Your two products (individual)

Everyone works on their own products. There are no groups.

1. **A research proposal on your own question**, from your own department, built one part at a time as homework and submitted in Session 8 (written proposal + 6 slides). About 5 volunteers also present their proposal in Session 8 (6 minutes + 2 minutes of feedback). Presenting is optional. Details: the proposal brief (`Proposal_Brief.pdf`, sent by email before Session 1).
2. **The Results section of a paper from the course dataset.** At the end of Session 6 you choose ONE of the five FLUID-ICU questions, Q1 to Q5. We give you the results in a results pack, and you write a submission-ready Results section. All sections of a paper are taught in Sessions 7 and 8. Only the Results section must be submitted, the others are optional. The instructor models the Results in Session 7, and you finish yours by 23:59 on Sunday 20 December 2026 (Vietnam time). After the course you review one classmate's draft (reviewer assigned at random in Session 7) and revise yours. Details: the manuscript brief (`Manuscript_Brief.pdf`, sent by email at the end of Session 6).

Between sessions (about 30 to 60 minutes a week)

After	Do this
Session 1	Draft your topic, PICO, research question, FINER check, aim and objectives. Think of synonyms for your PICO terms
Session 2	Finish Proposal Part 1 and the literature matrix. Read the abstract of Paper 2

After	Do this
Session 3	Draft Proposal Part 2 (design, setting, population)
Session 4	Finish Proposal Part 2
Session 5	Draft the first half of Proposal Part 3 (variables, CRF, data plan). Look at the five results packs (Q1 to Q5)
Session 6	Finish Proposal Part 3. Read the results pack of the question you chose. Start your 6 slides
Session 7	Finish Proposal Part 4 and your 6 slides, and email both before 19:30 on Saturday 19 December (Session 8). Write the Results section for your question
Session 8	Email your Results section to your reviewer and the instructor by 23:59 on Sunday 20 December. Send your review by Wednesday 30 December. Send your revision and response by Saturday 9 January 2027

What you need

A computer (or tablet) with a stable internet connection, camera and microphone (a headset helps), the meeting link sent before Session 1, and a quiet place for 3 hours. A pen and a notebook. No statistics software to install. Exercise sheets and other handouts are sent by email before each session. Participants who have done the free Neudata course *Clinical Data Analysis in R* may ask for the R scripts and re-run the course analyses.

Your toolkit

Every file comes by email, in English and Vietnamese (the Vietnamese file ends in `_VN`).

When	Files
Before Session 1	Proposal brief (`Proposal_Brief.pdf`), `Presentation_Guide.pdf`, `Proposal_Rubric.pdf`, the proposal template (`Research_Proposal_Template.docx`), Papers 1 and 2 (original PDFs), the Session 1 exercise sheet, the English–Vietnamese glossary (`Glossary_EN_VN.pdf`)
Before Session 2	`Literature_Review_Worksheet.docx`
Session 4	`Risk_of_Bias_Tool_Chooser.pdf` (before), `2x2_calculator.xlsx` (after)
Before Session 5	The five results packs (`Results_Pack_Q1.pdf` to `Results_Pack_Q5.pdf`), `Data_Dictionary.pdf`, `CRF_Template.docx`
Before Session 6	The FLUID-ICU analysis plan (`SAP_FLUID-ICU.pdf`), `Statistical_Analysis_Plan_Template.docx`, `sample_size_calculator.xlsx`
End of Session 6	Manuscript brief (`Manuscript_Brief.pdf`), `Manuscript_Template_IMRaD.docx`, `Manuscript_Rubric.pdf`, `Peer_Review_Instructions.pdf`, `Response_to_Reviewers_Template.docx`
Before Session 7	`Informed_Consent_Template.docx` and the list of reviewers

When	Files
Before Session 8	`Critical_Appraisal_Checklist.pdf`, `Writing_and_Submission_Checklist.pdf`
On request	Study protocol, cover letter and conference abstract templates, a sample ethics application and grant concept note (fictional), further reading, the R scripts of the course dataset

Every exercise sheet arrives by email before its session, as a Word file and a PDF.

The quiz

In Session 1 and again at the end of Session 8 you answer the same 24 questions online (a link is posted in the chat). You give your full name so that your two attempts can be compared. Only the course team sees names and scores, and you do not see the answers until after Session 8. The quiz measures what the *course* achieved, not you.

After the course

Take your proposal to the ethics committee. If you like, add the other sections to your Results section. This course is Level 1. If you want to run analyses yourself, Level 2 is a hands-on course in R (prerequisite: this course and the free *Clinical Data Analysis in R* course). Level 3 offers advanced, specialised methods on request.

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